

Lab 04 Objects and Classes

Objective

In this lab, you will learn how to:

- Clarifying the concepts of classes, objects, methods and instance variables.
- Declare a class and use it to create an object.
- Declare methods in a class to implement the class's behaviors.
- Call an object's method to make that method perform its task.
- Accessing and changing the state of an object using (.) operator.
- Creating overloaded constructors to ensure that an object's data is initialized when the object is created.

Activity Description

(The **Rectangle** class)

Design a class named **Rectangle** to calculate the area and perimeter of a rectangle.

Declare a class **Rectangle** with the following attributes:

- **Double** data field named **width** that specify the width of the rectangle.
- **Double** data field named **height** that specify the height of the rectangle.
The **default values** are **1** for both **width** and **height**.
- The class **Rectangle** must have the following **constructors**:
 - A no-arg constructor that creates a default rectangle.
 - A constructor that creates a rectangle with the specified **width** and **height**.
- The class **Rectangle** must have the following **behaviors**:
 - Two **setting methods** for width and height.
 - Two **getting methods** for width and height.
 - A method named **getArea()** that returns the area of this rectangle.
 - A method named **getPerimeter()** that returns the perimeter of this rectangle.
 - A method named **toString()** that return the rectangle information.

Implement class **RectangleTest** that demonstrates class **Rectangle**'s capabilities.

Write a test program that creates two **Rectangle** objects:

1. **Rectangle** with width **4** and height **40**
2. **Rectangle** with width **3.5** and height **35.9**.

Display the width, height, area, and perimeter of each rectangle in this order. (as shown in the output sample)

Sample Output:

```
run:
Rectangle No. 1
width = 4.0
Hight = 40.0
Area = 160.0
Perimeter = 88.0

Rectangle No. 2
width = 3.5
Hight = 35.9
Area = 125.64999999999999
Perimeter = 78.8

BUILD SUCCESSFUL (total time: 3 seconds)
```